Practice Problem Solutions

CM = Compulsory Miss
Miss = Conflict / Capacity

a) (C= 7, B =4, S=0) => 8 sets with one block in each set

1. 0xA1D4	CM
2. 0x31A7	CM
3. 0xA1E6	CM
4. 0xFF44	CM
5. 0xFF58	CM
6. 0x112D	CM
7. 0x1132	CM
8. 0xC1D0	CM
9. 0xFF78	CM
10. 0x31D2	CM
11. 0x1110	CM
12. 0x1126	Hit
13. 0x10D0	CM
14. 0x01E2	CM
15. 0xFF78	Hit
16. 0x113A	Hit
17. 0xA1D6	Miss
18. 0xA1DA	Hit
19. 0x1126	Hit
20. 0x10D5	Miss

Total Hits = 5
Total Misses = 15
Miss Rate = 0.75

b) (C= 7, B =4, S=3) => 1 set with 8 blocks

1. 0xA1D4	CM
2. 0x31A7	CM
3. 0xA1E6	CM
4. 0xFF44	CM
5. 0xFF58	CM
6. 0x112D	CM
7. 0x1132	CM
8. 0xC1D0	CM
9. 0xFF78	CM
10. 0x31D2	CM
11. 0x1110	CM
12. 0x1126	Hit

13. 0x10D0 CM
14. 0x01E2 CM
15. 0xFF78 Hit
16. 0x113A Hit
17. 0xA1D6 Miss
18. 0xA1DA Hit
19. 0x1126 Hit
20. 0x10D5 Hit

Total Hits = 6
Total Misses = 14
Miss Rate = 0.70

c) (C= 7, B =4, S=1) => 4 sets with 2 blocks

1. 0xA1D4 CM
2. 0x31A7 CM
3. 0xA1E6 CM
4. 0xFF44 CM
5. 0xFF58 CM
6. 0x112D CM
7. 0x1132 CM
8. 0xC1D0 CM
9. 0xFF78 CM
10. 0x31D2 CM
11. 0x1110 CM
12. 0x1126 Hit
13. 0x10D0 CM
14. 0x01E2 CM
15. 0xFF78 Hit
16. 0x113A Hit
17. 0xA1D6 Miss
18. 0xA1DA Hit
19. 0x1126 Hit
20. 0x10D5 Hit

Total Hits = 6
Total Misses = 14
Miss Rate = 0.70